

Experiment No. 4

Strategize impedance matching with characteristic impedance and load impedance through the measurement of (a) location of the stub and (b) length of the stub

Test Case 1

Optimize impedance matching using single stub matching by finding the location and length of the stub for values of load impedance from $100\angle 30^\circ$ to $350\angle 30^\circ$, keeping the characteristic impedance $Z_0 = 60\ \Omega$ constant.

PROGRAM:

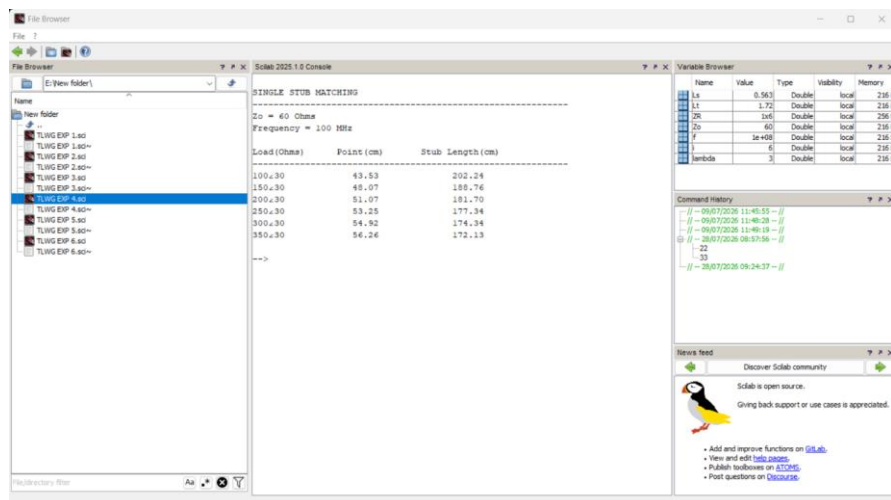
```
clear; clear; clc;
ZR = 100; Zo = 60; f = 100*(10^6);
lo = 300/(f*(10^-6)); // lo = wavelength
Ls = (lo/(2*pi))*(atan(sqrt(ZR/Zo)));
printf("-Point of attachment = %f cms\n", round(Ls*(10^2)*10)/10)
Lt = (lo/(2*pi))*(%pi + (atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf("-Length of the short circuited stub = %f cms", round(Lt*(10^2)));
```

OUTPUT:

Variable: ZR Constants: $Z_0 = 60\ \Omega$, $f = 100 \times 10^6\ \text{Hz}$

ZR (Ω , $\angle 30^\circ$)	Point of attachment (cm)	Length of short-circuited stub (cm)
100	43.53	202.24
150	48.07	188.76
200	51.07	181.70
250	53.25	177.34
300	54.92	174.34
350	56.26	172.13

OUTPUT SCREENSHOT:



Test Case 2

Optimize impedance matching using single stub matching by finding the location and length of the stub for values of characteristic impedance ranging from 600 Ω to 1800 Ω , keeping the load impedance $Z_R = 100\angle 30^\circ$ constant.

PROGRAM:

```
clear; clear; clc;
ZR = 100; Zo = 600; f = 100*(10^6);
lo = 300/(f*(10^-6)); // lo = wavelength
Ls = (lo/(2*pi))*(atan(sqrt(ZR/Zo)));
printf("-Point of attachment = %f cms\n", round(Ls*(10^2)*10)/10)
Lt = (lo/(2*pi))*(%pi + (atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf("-Length of the short circuited stub = %f cms", round(Lt*(10^2)));
```

OUTPUT:

Variable: Zo Constants: $Z_R = 100\angle 30^\circ \Omega$, $f = 100 \times 10^6$ Hz

Zo (Ω)	Point of attachment (cm)	Length of short-circuited stub (cm)
600	18.51	128.25
900	15.36	132.87
1200	13.42	135.43
1500	12.06	137.11
1800	11.05	138.32

OUTPUT SCREENSHOT:

The screenshot displays the Scilab 2025.1.0 interface. The main window shows the console output of the program, which calculates the point of attachment and the length of the short-circuited stub for various characteristic impedances (Zo) ranging from 600 to 1800 Ohms. The output is as follows:

Zo (Ohms)	Point (cm)	Stub Length (cm)
600	18.51	128.25
900	15.36	132.87
1200	13.42	135.43
1500	12.06	137.11
1800	11.05	138.32

The Variable Browser on the right shows the values of the variables defined in the program:

Name	Value	Type	Visibility	Memory
Ls	0.111	Double	local	216 B
Lt	1.38	Double	local	216 B
ZR	100	Double	local	216 B
Zo	1x5	Double	local	248 B
f	1e+08	Double	local	216 B
lambda	3	Double	local	216 B

The Command History shows the execution of the program, and the News feed displays a message about Scilab being open source.